A 3×4 ordered binary removal obstruction

October 11, 2026. **Candidate result with a written all-depth proof.** The exact finite checks use two implementations. A separate agent reviewed the argument; this is not independent mathematical peer review. This new host and main theorem are now checked in Lean, including the full quantifiers. See the source-bound audit and verification scope in Section 8.

The tree, threshold entries, mode roles, and arbitrary-repair sampler are inherited from OpenAI's September 25 manuscript. Erebos previously proved the 4×4 obstruction and exact repair equality. Our construction changes the anchor encoding, retaining the globally ordered modes from the preceding 4×4 quantitative refinement. The new ingredient is one anchor row with nonuniform extra ones, described explicitly below.

Statement

The fixed rectangular pattern is

$$H = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix}, \qquad P = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

For every integer h≥1, set m=2^h, d=14h+2, n=dm. The square binary host A_h below satisfies

$$\mathrm{dist}_H(A_h) \geq d^{-2}, \qquad N_H(A_h) \leq 2mn^5, \qquad \frac{N_H(A_h)}{n^7} \leq 2d^{-2}2^{-h}.$$

Copies use three increasing distinct rows and four increasing distinct columns and match all twelve entries. Distance allows arbitrary binary changes in either direction anywhere in the host, normalized by n^2. No equality for repair distance or exact all-depth copy count is claimed.

Consequently, for all c,C>0 and every ϵ_0>0, there exists $0<\epsilon<\epsilon_0$ such that hosts of arbitrarily large order are ϵ-far from H-freeness and have fewer than c ϵ^C n^7 copies. In particular H admits no polynomial removal bound. This reduces the pattern from the known 4×4 to 3×4 (16 to 12 entries); it does not establish a 3×3 obstruction or minimality.

1. The host, including every binary entry

Use a complete binary tree of height h, with nodes p=0,...,2^i−1 at depth i. For i<h put row blocks R_i^−(p),R_i^+(p) and column blocks C_i^−(p),C_i^+(p), each of size m/2^i. At depth h identify the two signs: R_h^−(p)=R_h^+(p)=L_R(p) and C_h^−(p)=C_h^+(p)=L_C(p), each of size one. Add dummy blocks R_,C_ of size m. A class omits the node index and denotes the union over its level; a parity subscript restricts node parity.

Rows follow recursive order: minus block, left subtree, right subtree, plus block. Columns follow plus block, left subtree, right subtree, minus block. A leaf block occurs once. Dummy rows are last, dummy columns first among nonanchors. Positions within a block are twins.

Every nonanchor entry meeting a dummy is one. All variable entries are zero except the following threshold entries:

Row / column	Value	Range
R_i^+(p), C_i^+(q)	1[p≤q]	0≤i≤h
R_i^−(p), C_i^−(q)	1[p<q]	0≤i<h

Row / column	Value	Range
R_i^+(p), C_{i−1}^+(q)	1[⌊p/2⌋≤q]	1≤i≤h
R_i^−(p), C_{i−1}^−(q)	1[⌊p/2⌋<q]	1≤i≤h

The last two rules both apply to a leaf row, with different column signs. There is no conflicting minus diagonal rule at leaves.

There are six modes at each i=1,. . . ,h, with these disjoint roles:

Mode	Row role 1	Row role 2	Column role 1	Column role 2
V_i^+	R_i^+	R_{i−1}^+	C_*	C_{i−1}^+
V_i^−	R_{i−1}^−	R_i^−	C_*	C_{i−1}^−
W_{i,d}^+	R_{i,d}^+	R_*	C_{i−1}^+	C_{i,d}^+
W_{i,d}^−	R_{i,d}^−	R_*	C_{i,d}^−	C_{i−1}^−

For each mode t add **one row anchor block X_t of size 2m** and **two column anchor blocks Y_{t,0},Y_{t,1}, each of size m**. Put anchors before all nonanchors. On both axes order modes as

$$(W^-_{1,0}, W^-_{1,1}, \ldots, W^-_{h,0}, W^-_{h,1}), \quad (W^+_{1,0}, W^+_{1,1}, \ldots, W^+_{h,0}, W^+_{h,1}), \quad (V^+_1, V^-_1, \ldots, V^+_h, V^-_h).$$

Within each column mode, group 0 precedes group 1. Anchor row X_t is one on Y_{t,0}, zero on every other anchor column. A nonanchor row in row role j∈{1,2} of t has signature e_j against (Y_{t,0},Y_{t,1}); a row in neither role has signature 00.

It remains to specify each anchor row on nonanchor columns. Write D=C_*. Its one-set T_t is exactly:

Anchor row mode	T_t
V_i^σ	C_{i−1}^σ
W_{i,d}^+	D ∪ C_{i,d}^+
W_{i,1}^−	D ∪ C_{i−1}^−
W_{i,0}^−, i<h	D ∪ C_{i−1}^− ∪ C_{i,1}^−
W_{h,0}^−	D ∪ C_{h−1}^−

In particular the odd-child addition is **absent at the common leaves**. All unspecified entries are zero. The added dummy and odd-child ones are outside the designated column roles. On those roles each X_t still has signature (0,1), as required by H's last two columns. Together with the prescribed row signatures, every ordered designated P extends to H.

Each axis has 12hm anchor positions and (2h+2)m nonanchor positions. Thus the host is square with n=(14h+2)m.

2. Elementary support and order facts

At a fixed signed level, blocks occur in increasing node order. For a fixed nonanchor row, entries along a column class are nondecreasing; for a fixed nonanchor column, entries along a row class are nonincreasing. Thus same-class rows or columns have nested neighborhoods.

Nonanchor rows and columns contain no unordered 3×3 crown J_3-I_3. Here is the inherited argument. Same-class columns cannot occur in a crown. A nondummy nonleaf row is supported on its own and preceding signed column classes. A leaf row is supported on L_C,C_{h−1}^+,C_{h−1}^−. Hence three distinct classes with pairwise common nondummy neighbors must be those three terminal classes. On fixed columns from those classes, leaf traces are nested as the leaf node increases, and nonleaf rows have zero on L_C. They cannot provide both required incomparable traces with a one on L_C.

The one-set of any anchor column on nonanchor rows is a whole class or a parity half-class. Any two such sets are disjoint or nested. No mode order assumption is needed for this laminarity statement.

We repeatedly use the recursive-order comparisons, with $\pi(p)=\lfloor p/2 \rfloor$:

$$\begin{aligned} C^+_{i-1}(b) < C^\sigma_i(q) &\iff b \le \pi(q), & C^\sigma_i(q) < C^+_{i-1}(b) &\iff \pi(q) < b,\\ C^\sigma_i(q) < C^-_{i-1}(b) &\iff \pi(q) \le b, & C^-_{i-1}(b) < C^\sigma_i(q) &\iff b < \pi(q),\\ C^+_j(a) < C^-_j(b) &\iff a \le b, & C^-_j(a) < C^+_j(b) &\iff a < b,\\ R^+_i(p) < R^+_{i-1}(a) &\iff \pi(p) \le a, & R^-_{i-1}(a) < R^-_i(p) &\iff a \le \pi(p). \end{aligned}$$

The two same-level cross-sign comparisons require j<h; at depth h the signs are identified. Child signs in the parent/child comparisons may be the common leaf sign. Reversing a strict block comparison gives its complementary strict inequality. These facts follow directly from the consecutive left-to-right subtree intervals.

3. Exactly one anchor row and one or two anchor columns

Consider a copy on rows $r_1<r_2<r_3$ and columns $c_1<c_2<c_3<c_4$. Let a,b count the selected anchors on each axis. Anchors are prefixes. H has three distinct rows and four distinct columns, so no twin block can be reused.

No anchor rows (a=0). If b=0, columns 1,3,4 form a crown, impossible. If b=1 or 2, column 1 places rows 1 and 2 in one class, but their traces on nonanchor columns 3,4 are 01 and 10. If b≥3, anchor columns 1 and 3 have traces 110 and 011, contradicting laminarity.

At least two anchor rows with b>0. Column 1 is one on the first two anchor rows. It can only meet one mode's row block, so those rows would be twins, impossible. Hence a≥2 requires b=0.

For the next cases let A<B be the modes of the first two anchor rows. Their supports must have a common c_1 and exclusive columns $c_3 \in T_B \setminus T_A$, $c_4 \in T_A \setminus T_B$. If c_1 is variable, c_4 cannot be dummy since $c_1<c_4$. Inspection of the explicit support table leaves precisely

$$A = W^-_{i,0}, \quad i < h, \qquad B = W^-_{i+1,d} \text{ or } V^-_{i+1}.$$

Indeed the only other crossing pair with variable intersection is W_{i,d}^+,V_{i+1}^+, whose first-only part is dummy. Same-level minus supports are nested, disjoint-level supports have no variable intersection, and vertical supports are single distinct classes. For the displayed pairs, c_1 is in odd C_i^−, c_4 in C_{i−1}^−, and c_3 in even C_i^−, or possibly odd C_{i+1}^− when B=W^-_{i+1,0} with i+1<h.

Two anchor rows (a=2,b=0). Column c_1 is variable since its bottom entry is zero. In the displayed pair classification, a nondummy row cannot be one on both C_{i−1}^− and C_{i+1}^−. Therefore c_3 is in even C_i^− and r_3 must be in R_i^−. Its entire nonanchor one-support lies in D∪C_{i−1}^−∪C_i^−, covered by T_A∪T_B. This contradicts c_2's trace 001.

Three anchor rows (a=3,b=0). If c_1 is variable, the same pair classification applies. No anchor row is one both on $c_4 \in C_\{i-1\}\hat{}-$ and c_3 in even $C_i\hat{}-$ or odd $C_\{i+1\}\hat{}-$: the sole support spanning the adjacent minus classes adds only odd children. If c_1 is dummy, its trace 110 forces the first two modes to be W and the last to be V. The V support is one full class K, so c_2,c_3,c_4 all lie in K. A W support restricted to K is empty, all of K, or one parity half. The exclusive c_3,c_4 force opposite halves, whose union covers K; there is no c_2 with trace 001.

It follows that a=1. If b=4, c_1 and c_4 must both be the sole group-0 anchor class on which r_1 is one; these are twins, impossible. If b=3, column c_3 places r_2,r_3 in one row class, whereas the nonanchor c_4 has trace 01 on them, contradicting nonincreasing same-class row entries.

No anchor columns (a=1,b=0). Column c_1 is not dummy, since its bottom entry is zero; c_3 is not dummy either, since $c_1<c_2<c_3$. Both lower rows are nondummy and lie in different classes, since their column traces include both 10 and 01. Their common one at c_3 forces adjacent signed row classes, possibly including the terminal leaf class.

For a V mode, a W$\hat{}+$ mode, or an unextended W$\hat{}-$ mode, c_1,c_4 lie in one column class. Their order contradicts $r_2 \cdot c_1=1$,$r_2 \cdot c_4=0$. The sole remaining mode is W$\hat{}$-*{i,0}, i<h. Put* $P_0=C\{i-1\}\hat{}-$ and Q_0=odd $C_i\hat{}-$; c_1,c_4 must lie in different classes.

If $c_1 \in P_0$ and $c_4 \in Q_0$, the lower row classes are either $(R_\{i-1\}\hat{}-,R_i\hat{}-)$, forcing their common c_3 into P_0 where its top entry is one, or $(R_i\hat{}-,R_\{i+1\}\hat{}-)$. In the latter case, write $c_1=P_0(b)$, $c_4=Q_0(q)$, $r_2=R_i\hat{}-(p)$. Column order gives $b<\pi(q)$, while $r_2 \cdot c_1=1$ gives $\pi(p)<b$. Hence $p<q$, contradicting $r_2 \cdot c_4=0$.

If $c_1 \in Q_0$ and $c_4 \in P_0$, the lower row classes are either $(R_i\hat{}-,R_\{i-1\}\hat{}-)$, again forcing c_3 into P_0, or $(R_\{i+1\}\hat{}-,R_i\hat{}-)$. In the latter case let r_2 have node p and r_3 node z. Since c_2 has top zero and bottom one, it must be an even $C_i\hat{}-$ column with node u: dummy and P_0 are in the top support. Row order gives $\pi(p)<z$; the bottom entry gives $z<u$. Thus $r_2 \cdot c_2=1$, contradicting its required zero. If i+1=h, the same minus incidences and row-order comparison hold with the common leaf class. This finishes the b=0 exclusion.

We have proved $(a,b) \in \{(1,1),(1,2)\}$.

4. Every copy uses its own terminal minus parent

Let t be r_1's mode. Column c_1 is $Y_\{t,0\}$, so r_2 belongs to row role 1 of t. The remaining conditions include

$$c_3 < c_4, \quad A(r_2, c_3) = 1, \quad A(r_2, c_4) = 0, \quad c_3 \notin T_t, \quad c_4 \in T_t.$$

Both c_3,c_4 are nonanchors. Column c_4 is not dummy, since $c_3<c_4$.

Vertical t. For $V_i\hat{}+$, r_2 is $R_i\hat{}+(p)$ and $c_4=C_\{i-1\}\hat{}+(q)$ with $\pi(p)>q$. A nondummy c_3 outside T_t with a one on r_2 would be in $C_i\hat{}+$ (or, at a leaf, $C_\{h-1\}\hat{}-$). Its order before c_4 contradicts the threshold inequalities. Explicitly a $C_i\hat{}+$ node u requires $p \le u$ and $\pi(u)<q$; a terminal minus-parent node u requires $\pi(p)<u<q$. For $V_i\hat{}-$, r_2 is $R_\{i-1\}\hat{}-(p)$ and $c_4=C_\{i-1\}\hat{}-(q)$ with $p \ge q$. A nondummy c_3 outside T_t could only be $C_\{i-2\}\hat{}-(u)$ when $i \ge 2$, requiring $\pi(p)<u<\pi(q)$, also impossible. At i=1 that preceding class is absent. Thus c_3 must be dummy.

When b=1, the nonanchor $c_2<c_3$ rules that out. When b=2, a later nondummy r_3 with a one on c_4 is impossible by the same threshold and row-order inequalities: in the plus case its row class is $R_\{i-1\}\hat{}+$ or $R_i\hat{}+$, and in the minus case $R_\{i-1\}\hat{}-$ or $R_i\hat{}-$. Hence r_3 is dummy. Its one at anchor c_2 would require a W group-1 column. Every W anchor precedes the V column c_1, contradicting $c_1<c_2$.

Positive horizontal t=W_{i,d}ˆ+. Column c_4 is in C_{i,d}ˆ+, as is the class of r_2. Write their nodes p,q. The zero requires p>q. A c_3 in the same child class is excluded by monotonicity and $c_3<c_4$. A plus parent c_3=C_{i−1}ˆ+(u) requires $\pi(p)\le u\le\pi(q)$. Equal floors would make p,q unequal siblings, contradicting their common parity d. At the terminal level a minus-parent c_3 instead requires $\pi(p)<u<\pi(q)$, impossible. The dummy is in T_t and cannot be c_3. No copy occurs in this mode.

Negative horizontal t=W_{i,d}ˆ−. The row r_2 is R_{i,d}ˆ−(p), or L_R(p) if i=h. If c_4 is an added odd child, then i<h,d=0. Its node q satisfies p≥q. Column c_3 outside T_t with a one on r_2 can only be an own-level node u>p; $c_3<c_4$ would give u<q, impossible.

Thus c_4=C_{i−1}ˆ−(b), with $\pi(p)\ge b$. For i<h, a c_3 outside T_t with a one on r_2 must be C_iˆ−(q), where q>p and $\pi(q)\le b$. These inequalities force $\pi(q)=b=\pi(p)$, p even and q=p+1 odd. The row role then has d=0, but this odd c_3 was added to T_t, a contradiction. This is the precise use of the new odd-child ones.

Therefore i=h. Here c_3 can only be a leaf column q with p≤q, or a plus-parent column C_{h−1}ˆ+(a) with $\pi(p)\le a$. In the first case its order before c_4 gives $\pi(q)\le b$; in the second it gives a≤b. Together with $\pi(p)\ge b$, both cases imply $b=\pi(p)$.

Every copy consequently sends its fixed zero H(2,4) to

$$T_h = \{(L_R(p), c) : c \in C^-_{h-1}(\lfloor p/2 \rfloor)\}.$$

There are exactly 2m cells in T_h, because each terminal parent block has two positions. Fixing this cell leaves at most n^5 choices for the other two rows and three columns. Thus N_H(A_h)≤$2mn^5$.

5. Arbitrary repairs still cost at least m^2

Let a binary B differ from A_h on M<m^2 cells. Sample a uniform leaf z and its ancestor at each level. Independently choose one uniform position from each signed row and column block on this path, identifying the two sign selections at the leaf. Choose independent uniform representatives of each anchor block and each dummy block.

A variable or dummy position has marginal probability 1/m. A column anchor position has probability 1/m, and a row anchor position 1/(2m). Selections with an anchor or dummy endpoint are independent across axes; root selections are also independent. Therefore any protected cell (anchor endpoint, dummy endpoint, or root-to-root) is selected with probability at most $1/m^2$. The union bound supplies a selection with no edited protected cell. This probability assertion is not made for arbitrary variable-variable cells or for shared leaf diagonals.

Suppose B is H-free. The unchanged designated anchors extend every ordered P body on the selected role representatives to H. Therefore all those bodies must avoid P. The root-plus diagonal is one and the root-minus diagonal zero. If the selected parent diagonals have these values, the vertical bodies are

$$\begin{pmatrix} 1 & b_+ \\ 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ 1 & b_- \end{pmatrix}.$$

Their P-freeness forces b_+=1,b_−=0. The horizontal modes matching the child parity next have bodies

$$\begin{pmatrix} 1 & c_+ \\ 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} c_- & 0 \\ 1 & 1 \end{pmatrix},$$

forcing child diagonal values c_+=1,c_−=0. The recursive axis orders put all these role bodies in increasing order. Induction reaches the common leaf diagonal, which cannot simultaneously be one and zero. Thus B is not H-free. All M edits were arbitrary, including anchor edits and both directions of binary change. This proves the m^2 lower bound.

6. Quantifiers and comparison

With ϵ_h=(14h+2)^−2, the density-to-power ratio is at most

$$\frac{N_H(A_h)}{n_h^7 \epsilon_h^C} \leq 2(14h+2)^{2C-2} 2^{-h} \longrightarrow 0.$$

Given c,C,ϵ_0>0, choose h with ϵ_h<ϵ_0 and this ratio<c. Replace every physical row and column by q identical consecutive twins, for any positive integer q. The same sampler gives repair lower bound q^2m^2. Since H has distinct rows and columns, a copy cannot reuse a projected row or column, so its count scales exactly by q^7. Both normalized bounds are unchanged. Choosing q arbitrarily large gives the required host sizes, including a prescribed threshold depending on ϵ_h.

The improvement is rectangular pattern size: 3×4 rather than the known 4×4. Copy densities have different normalizations n^7 and n^8, so the raw count is not a multiplicative improvement to the old fixed-H bound. The earlier 4×4 exact-distance and optimized-count results remain separate. Neither the exact repair equality nor its existing Lean certificate has been transferred to this host.

7. Reproduction and verification scope

`experiments/asymmetric_candidate.py` enumerates all ordered quotient copies with exact twin multiplicities through h=6. `verification/check_three_by_four.py` separately constructs the host and uses weighted subsequence dynamic programming. It checks the same depths, a fully expanded h=1 matrix, designated anchor signatures, and adverse variants omitting each of the two anchor-row changes. The latter variants have nonlocalized copies. See the archived exact report and separate agent review. These finite computations support this written argument; they do not supply its universal quantifiers or independent peer review.

8. Lean formalization

The main theorem is checked in Lean 4.34.1 with the pinned Mathlib dependency.

`no_polynomial_removal_bound` in the `OrderedRemoval.Rectangular` namespace states the full quantifier order: for every c,C,ϵ_0>0, choose a fixed 0<ϵ<ϵ_0; for every host threshold N, there is an n≥max(1,N) square binary matrix with at least ϵn^2 edits in every H-free repair and fewer than c ϵ^C n^7 copies.

The proof constructs the literal host and both orders, excludes every unwanted anchor configuration, localizes the fixed zero, counts copies, and proves the arbitrary-repair bound before normalizing. The uniform-blowup lemmas preserve the normalized bounds. They prove the sufficient q^7 copy upper bound; the exact blowup-count equality in Section 6 is not needed by the formal capstone. No exact all-depth enumeration or repair equality is asserted for this host.

The formal sampler is an alternative to Section 5's independent sampler: it retains the inherited three seeds and selects representatives from both m-sized row subgroups of each 2m anchor block. Copy completion uses the second representative. Its anchor-row marginal is 1/m, and every protected cell still has probability at most $1/m^2$. This distinction changes no bound.

The audit binds all project source files. It checks transitive axiom dependencies and replays each project module into its imported Lean environment. This does not freshly replay all

Mathlib or constitute independent mathematical peer review. Only `propext`, `Quot.sound`, and `Classical.choice` occur as axioms. The separate-agent semantic review is archived alongside the finite checks.